

• APPLICATION OF DL •

# Image Segmentation  $\Rightarrow$  Image is divided into multiple segments to simplify its representation and make analysis easier.

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• Objectives  $\Rightarrow$

$\rightarrow$  Identify objects and boundaries.

$\rightarrow$  Simplify image representation.

$\rightarrow$  Enable object recognition.

$\rightarrow$  Extract useful feature for further processing.

• Types  $\Rightarrow$

1. Semantic Segmentation:

$\rightarrow$  assign a class label to each pixel.

$\rightarrow$  all objects of the same type share the same label, for e.g.

all cars = one category

2. Instance Segmentation:

$\rightarrow$  it detects individual objects separately.

$\rightarrow$  each object has a unique identity, for e.g. car1  $\neq$  car2.

3. Panoptic Segmentation:

$\rightarrow$  it is the combination of semantic and instance segmentation.

$\rightarrow$  provides both class and instance information

• Techniques used in Segmentation:

1. Thresholding  $\Rightarrow$  it is the simplest method and it can convert grayscale image into binary image, it is based on intensity

value. For e.g. if the pixel value is greater than threshold  $\rightarrow$  object else background.

2. Edge based segmentation  $\rightarrow$  it detects edges using operator like Sobel operator or Kenney Edge detection. It works well when object boundaries are clear.

3. Region based segmentation  $\rightarrow$  it divides image into regions based on similarity. It uses two methods i.e. region growing and region splitting and merging.

4. Clustering based segmentation  $\rightarrow$  in this method groups pixel based on similarity it uses algorithms like K-mean clustering and Mean shift.

## # Object Detection $\rightarrow$

$\rightarrow$  It is a key task in computer vision where a system identifies object in an image or videos and it locates them using bounding boxes.

### • Key Components:

1. Classification  $\rightarrow$  determines the class label of objects.

2. Localization  $\rightarrow$  find the position of objects using bounding boxes.

3. Bounding Box  $\rightarrow$  a rectangular box around the detected object, which include class label and confidence score.

### • Application:

1. Autonomous vehicle: detect cars, pedestrians, traffic signal.

2. Medical Imaging: detect tumors, abnormalities.

3. Security & Surveillance: face detection, intruder detection.

4. Retail: product detection, inventory tracking.

5. Mobile Apps: AR filters.

### • Advantages:

- $\rightarrow$  enable real time decision making.
- $\rightarrow$  works on images and videos.
- $\rightarrow$  foundation of many AI system.

### • Challenges:

- $\rightarrow$  it cannot detect small object.
- $\rightarrow$  Occlusion (object overlapping)
- $\rightarrow$  lighting variation.